Solving the Game

Finding MI6's Optimal Strategy (p_t, p_f) by the Corner Method

1 What Kind of Equilibrium Is This?

It is tempting to call this a *Nash equilibrium* problem, since MI6 and Cheka are clearly playing against one another. But a Nash equilibrium requires **both** players to be choosing a strategy, with each strategy being a best response to the other. That is not the situation here.

Cheka has no strategic freedom in this version of the game. Cheka is told p_t exactly, and it is committed, by construction, to two mechanical steps:

1. Form the exact Bayesian posterior over the three cities, using the known p_t.
2. Search the city with the **maximum** posterior probability.

There is no room in this rule for Cheka to "decide" anything — given a message, its search action is a fixed, deterministic function. Cheka is not optimizing; it is simply computing.

MI6, on the other hand, has two genuine free parameters: p_t (how often to tell the truth) and p_f (how often to follow its own report). MI6 can choose these to shape Cheka's belief however it likes, *because MI6 knows in advance exactly how Cheka will react to any message it sends.*

So what are we actually solving?

This is a **one-sided constrained optimization** — closer to a *Stackelberg* setup (a leader who moves first and knows the follower's exact reaction function) than to a Nash equilibrium. We are finding:

$$(p_t^*,\, p_f^*) \;=\; \underset{p_t,\, p_f \,\in\, [0,1]}{\arg\max}\;\; \mathbb{E}[\text{MI6's score} \mid p_t, p_f, r]$$

subject to Cheka's response being the fixed rule above. It ensures MI6 cannot do any better *given that Cheka will always behave exactly this way.* We are not claiming Cheka is playing optimally for itself; we take its rule as given and ask what the best counter-strategy is.

2 Cheka's Decision Rule

Cheka starts with a uniform prior ($\frac{1}{3}$ each). Upon receiving message m, its posterior is built from the likelihood

$$\text{likelihood}(m) = p_t, \qquad \text{likelihood(other two)} = \frac{1-p_t}{2} \text{ each.}$$

The prior is uniform, so it cancels out of Bayes' rule, and the posterior equals the likelihood directly:

$$\text{posterior}(m) = p_t, \qquad \text{posterior(each other city)} = \frac{1-p_t}{2}.$$

2.1 The Trust Threshold

Cheka searches the message location iff its posterior there exceeds the alternatives:

$$\begin{aligned} p_t &> \frac{1-p_t}{2} \\ 2p_t &> 1-p_t && \text{(multiply both sides by 2)} \\ 3p_t &> 1 && \text{(add } p_t \text{ to both sides)} \\ p_t &> \frac{1}{3} \end{aligned}$$

This splits the game into two regimes, each with its own p_t-domain:

- **Trust regime** ($p_t \in [\frac{1}{3}, 1]$): Cheka always searches *the message location*.
- **Distrust regime** ($p_t \in [0, \frac{1}{3}]$): Cheka always searches *one of the other two*, uniformly.

(We include the boundary $p_t = \frac{1}{3}$ in both, since — as we'll confirm — the two formulas agree exactly there.)

3 Setting Up MI6's Expected Score

Since "both find it" pays MI6 only 3 (instead of 10), we can write MI6's expected score compactly as

$$\mathbb{E}[\text{MI6}] \;=\; 10\,\underbrace{P(\text{MI6 finds it})}_{\mu} \;-\; 7\,\underbrace{P(\text{both find it})}_{P(\text{both})},$$

since $10\,P(\text{MI6 alone}) + 3\,P(\text{both}) = 10\big[P(\text{MI6 alone}) + P(\text{both})\big] - 7\,P(\text{both})$.

MI6's own accuracy follows directly from the definition of p_f:

$$\mu = P(\text{MI6 finds it}) = p_f \cdot r + (1-p_f)\cdot\frac{1-r}{2}.$$

$P(\text{both})$ requires tracking how MI6's search (driven by p_f, off the report R) and Cheka's search (driven by p_t, off the message M, itself derived from R) correlate through their shared root cause, the report R.

3.1 Trust Regime: $P(\text{both})$

We need $P(S_{\text{mi6}} = L,\ M = L)$. Condition on R, since S_{mi6} and M are independently randomized once R is fixed:

Case $R = L$ (probability r)

$P(S_{\text{mi6}} = L \mid R = L) = p_f$, $\quad P(M = L \mid R = L) = p_t$. $\quad$ Contributes: $r p_f p_t$.

Case $R \neq L$ (probability $1-r$, split over 2 wrong reports)

$P(S_{\text{mi6}} = L \mid R \neq L) = \frac{1-p_f}{2}$, $\quad P(M = L \mid R \neq L) = \frac{1-p_t}{2}$. $\quad$ Contributes: $(1-r)\dfrac{(1-p_f)(1-p_t)}{4}$.

$$\boxed{P(\text{both} \mid \text{trust}) \;=\; r\,p_f\,p_t \;+\; \frac{(1-r)(1-p_f)(1-p_t)}{4}}$$

3.2 Distrust Regime: $P(\text{both})$

Here S_{cheka} is uniform over the two cities $\neq M$, so $S_{\text{cheka}} = L$ needs *both* $M \neq L$ *and* the coin flip landing on L (probability $\frac{1}{2}$):

$$P(\text{both} \mid \text{distrust}) = \frac{1}{2} \sum_R P(R) P(S_{\text{mi6}} = L \mid R)\, P(M \neq L \mid R)$$

Case $R = L$

$P(S_{\text{mi6}} = L \mid R = L) = p_f$, $\quad P(M \neq L \mid R = L) = 1 - p_t$. $\quad$ Contributes: $r\,p_f(1-p_t)$.

Case $R \neq L$

$P(S_{\text{mi6}} = L \mid R \neq L) = \frac{1-p_f}{2}$, $\quad P(M \neq L \mid R \neq L) = \frac{1+p_t}{2}$. $\quad$ Contributes: $(1-r)\dfrac{(1-p_f)(1+p_t)}{4}$.

$$\boxed{P(\text{both} \mid \text{distrust}) \;=\; \frac{r\,p_f(1-p_t)}{2} \;+\; \frac{(1-r)(1-p_f)(1+p_t)}{8}}$$

(Both formulas were checked against direct Monte-Carlo simulation of the exact game and matched to 4 decimal places.)

3.3 MI6's Score in Each Regime

Substituting into $\mathbb{E}[\text{MI6}] = 10\mu - 7P(\text{both})$:

$$E_{\text{trust}}(p_t, p_f) = 10\left[p_f r + (1-p_f)\tfrac{1-r}{2}\right] - 7\left[r\,p_f\,p_t + \tfrac{(1-r)(1-p_f)(1-p_t)}{4}\right]$$

$$E_{\text{distrust}}(p_t, p_f) = 10\left[p_f r + (1-p_f)\tfrac{1-r}{2}\right] - 7\left[\tfrac{r\,p_f(1-p_t)}{2} + \tfrac{(1-r)(1-p_f)(1+p_t)}{8}\right]$$

4 A Cleaner Way to Optimize: the Corner Method

We now have two functions of *two* variables, $E(p_t, p_f)$, each defined on a rectangle, and we want their maximum (and, for completeness, their minimum).

4.1 The general recipe for two variables on a rectangle

For a continuous function $E(x, y)$ on a closed rectangle, extreme points can only occur in three places:

1. **Interior critical points:** solve $\dfrac{\partial E}{\partial x} = 0$ and $\dfrac{\partial E}{\partial y} = 0$ simultaneously, then classify with the Hessian.
2. **Edge interior points:** freeze one variable at its boundary value and optimize the resulting single-variable function of the other.

3. **Corners:** the four corner points of the rectangle.

In general, you must check all three. But our functions have a special structure that lets us skip straight to step 3.

4.2 E is bilinear — so only the corners matter

Expand E_{trust}:

$$E_{\text{trust}}(p_t, p_f) = -\tfrac{21}{4}p_f r p_t + \tfrac{53}{4}p_f r - \tfrac{7}{4}p_f p_t - \tfrac{13}{4}p_f - \tfrac{7}{4}r p_t - \tfrac{13}{4}r + \tfrac{7}{4}p_t + \tfrac{13}{4}$$

Notice: there is no p_t^2 term and no p_f^2 term — only a constant, linear terms in p_t and p_f, and a single cross term $p_t p_f$. A function of this shape is called **bilinear**. (The same is true of E_{distrust}.)

Why bilinear functions only need corners checked

Freeze p_f at any fixed value: E becomes a *straight line* in p_t (no curvature). A straight line has no interior maximum or minimum — its extreme values sit at the two endpoints of whatever interval p_t ranges over. The same is true freezing p_t and varying p_f. So:

- Step 1 (interior critical points) never contributes — there is no curvature to create one.
- Step 2 (edges) collapses immediately into step 3, since each edge is itself a straight line, whose own extremes are its two endpoints — i.e. two of the rectangle's corners.

Conclusion: both the maximum and the minimum of a bilinear function on a rectangle occur at one of its four corners. We only need to evaluate E at those four points and compare.

5 Applying the Corner Method — Trust Regime

Domain: $p_t \in \left[\frac{1}{3}, 1\right]$, $p_f \in [0, 1]$. The four corners are $\left(\frac{1}{3}, 0\right), \left(\frac{1}{3}, 1\right), (1, 0), (1, 1)$.

Corner $(p_t, p_f) = \left(\frac{1}{3}, 1\right)$

$$E_{\text{trust}}\left(\tfrac{1}{3}, 1\right) = 10\left[r + 0\right] - 7\left[r \cdot \tfrac{1}{3} + 0\right] = 10r - \tfrac{7r}{3} = \tfrac{30r - 7r}{3} = \boxed{\tfrac{23r}{3}}$$

Corner $(p_t, p_f) = (1, 0)$

$$E_{\text{trust}}(1, 0) = 10\left[0 + \tfrac{1-r}{2}\right] - 7\left[0 + \tfrac{(1-r)(1-1)}{4}\right] = 5(1-r) - 0 = \boxed{5(1-r)}$$

Corner $(p_t, p_f) = \left(\frac{1}{3}, 0\right)$

$$E_{\text{trust}}\left(\tfrac{1}{3}, 0\right) = 10\left[0 + \tfrac{1-r}{2}\right] - 7\left[0 + \frac{(1-r)\left(1-\frac{1}{3}\right)}{4}\right] = 5(1-r) - 7(1-r) \cdot \tfrac{2/3}{4}$$

$$= 5(1-r) - \tfrac{7(1-r)}{6} = \tfrac{30(1-r) - 7(1-r)}{6} = \boxed{\tfrac{23(1-r)}{6}}$$

Corner $(p_t, p_f) = (1, 1)$

$$E_{\text{trust}}(1,1) = 10[r+0] - 7[r \cdot 1 + 0] = 10r - 7r = \boxed{3r}$$

Corner (p_t, p_f)	E_{trust}
$\left(\frac{1}{3}, 1\right)$	$\dfrac{23r}{3}$
$\left(\frac{1}{3}, 0\right)$	$\dfrac{23(1-r)}{6}$
$(1, 0)$	$5(1-r)$
$(1, 1)$	$3r$

6 Applying the Corner Method — Distrust Regime

Domain: $p_t \in \left[0, \frac{1}{3}\right]$, $p_f \in [0, 1]$. The four corners are $(0,0), (0,1), \left(\frac{1}{3}, 0\right), \left(\frac{1}{3}, 1\right)$.

Corner $(p_t, p_f) = (0, 1)$

$$E_{\text{distrust}}(0,1) = 10[r+0] - 7\left[\frac{r \cdot 1 \cdot (1-0)}{2} + 0\right] = 10r - \frac{7r}{2} = \boxed{\frac{13r}{2}}$$

Corner $(p_t, p_f) = (0, 0)$

$$E_{\text{distrust}}(0,0) = 10\left[0 + \frac{1-r}{2}\right] - 7\left[0 + \frac{(1-r)(1+0)}{8}\right] = 5(1-r) - \frac{7(1-r)}{8} = \boxed{\frac{33(1-r)}{8}}$$

Corners at $p_t = \frac{1}{3}$ **— consistency check with the Trust regime**

$$E_{\text{distrust}}\left(\tfrac{1}{3}, 1\right) = 10r - 7\left[\frac{r\left(1-\frac{1}{3}\right)}{2}\right] = 10r - \frac{7r}{3} = \frac{23r}{3} \qquad \text{(same as trust regime!)}$$

$$E_{\text{distrust}}\left(\tfrac{1}{3}, 0\right) = 5(1-r) - 7\left[\frac{(1-r)\left(1+\frac{1}{3}\right)}{8}\right] = 5(1-r) - \frac{7(1-r)}{6} = \frac{23(1-r)}{6} \qquad \text{(same!)}$$

This confirms the two regimes' formulas meet *exactly* at their shared boundary $p_t = \frac{1}{3}$ — there is no jump, so the boundary corners are shared, not duplicated.

Corner (p_t, p_f)	E_{distrust}
$(0,1)$	$\dfrac{13r}{2}$
$(0,0)$	$\dfrac{33(1-r)}{8}$
$\left(\frac{1}{3},1\right)$	$\dfrac{23r}{3}$
$\left(\frac{1}{3},0\right)$	$\dfrac{23(1-r)}{6}$

7 Assembling the Full Picture: Six Candidates

Since the two boundary corners coincide, the two regimes together give exactly **six distinct candidate points** for the global max/min over the whole square $[0,1] \times [0,1]$:

Corner (p_t, p_f)	E
$(0,0)$	$\dfrac{33}{8}(1-r) = 4.125(1-r)$
$(0,1)$	$\dfrac{13}{2}r = 6.5r$
$\left(\frac{1}{3},0\right)$	$\dfrac{23}{6}(1-r) \approx 3.833(1-r)$
$\left(\frac{1}{3},1\right)$	$\dfrac{23}{3}r \approx 7.667r$
$(1,0)$	$5(1-r)$
$(1,1)$	$3r$

Each is a straight line in r. The global maximum (and minimum) of E, for any given r, is simply whichever line is highest (lowest) at that r — found by comparing all six.

8 Finding the Global Maximum

Evaluating all six lines across $r \in [0,1]$, only two of them ever reach the top: $(1,0)$ and $\left(\frac{1}{3},1\right)$. Find where they cross:

$$\begin{aligned}
5(1-r) &= \frac{23r}{3} \\
15(1-r) &= 23r \qquad \text{(multiply both sides by 3)} \\
15 - 15r &= 23r \\
15 &= 38r \\
r &= \frac{15}{38} \approx 0.395
\end{aligned}$$

Result — Global Maximum

$$r < \frac{15}{38}: \quad (p_t^*, p_f^*) = (1, 0), \quad E^* = 5(1-r)$$

$$r > \frac{15}{38}: \quad (p_t^*, p_f^*) = \left(\tfrac{1}{3}, 1\right), \quad E^* = \frac{23r}{3}$$

Example ($r = 0.6$): $\frac{23(0.6)}{3} = 4.6$. **Example ($r = 0.2$):** $5(0.8) = 4.0$.

Why this differs slightly from "$r > \frac{1}{3}$": Cheka's own trust threshold sits at r-independent $p_t = \frac{1}{3}$, but MI6's *optimal-strategy* crossover sits at $r = \frac{15}{38} \approx 0.395$ — a different number. In the narrow window $\frac{1}{3} < r < \frac{15}{38}$, Cheka is technically in "trust mode", yet MI6 still does better abandoning the $\left(\frac{1}{3}, 1\right)$ strategy in favor of full honesty with no self-following, $(1, 0)$. The corner method catches this automatically, because it compares *every* candidate at once rather than reasoning regime-by-regime.

9 Bonus: Finding the Global Minimum

The same six candidates immediately tell us the *worst* choices too — useful as a check that MI6's strategy design actually matters. Only two lines ever reach the bottom: $(1, 1)$ and $\left(\frac{1}{3}, 0\right)$.

$$\begin{aligned}
3r &= \frac{23(1-r)}{6} \\
18r &= 23(1-r) \qquad \text{(multiply both sides by 6)} \\
18r &= 23 - 23r \\
41r &= 23 \\
r &= \frac{23}{41} \approx 0.561
\end{aligned}$$

Result — Global Minimum (worst case for MI6)

$$r < \frac{23}{41}: \quad \text{worst } (p_t, p_f) = (1, 1), \quad E_{\min} = 3r$$

$$r > \frac{23}{41}: \quad \text{worst } (p_t, p_f) = \left(\tfrac{1}{3}, 0\right), \quad E_{\min} = \frac{23(1-r)}{6}$$

Notice $(1, 1)$ — full honesty *and* always following the report — is the single worst strategy for most values of r. This makes intuitive sense: at $p_t = p_f = 1$, MI6's message and MI6's search are *identical*, so Cheka (trusting the message) and MI6 end up searching the exact same city every time. Every success becomes a shared success (worth only 3, not 10) — MI6 gives away half its own reward for nothing.

10 Summary

Range of r	p_t^*	p_f^*	**MI6's Optimal Score**
$r < \frac{15}{38}$ (≈ 0.395)	1	0	$5(1-r)$
$r > \frac{15}{38}$	$\frac{1}{3}$	1	$\dfrac{23r}{3}$

Method recap. (1) Derive $E(p_t, p_f)$ for each of Cheka's regimes. (2) Notice E is bilinear — no p_t^2 or p_f^2 terms. (3) By the corner theorem, only the corners of each regime's rectangle can hold an extremum. (4) Evaluate E at all six distinct corners (the two regimes share a boundary). (5) Each corner value is a simple line in r; compare them to find which wins for a given r, and solve for the exact crossover algebraically. No derivatives were needed anywhere in the final optimization — only in confirming, once, that E has no curvature.